

1) Graph the function.

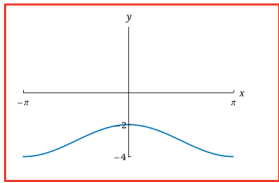
$$f(x) = -3 + \cos(x) \quad \text{HMLMH}$$

1) $A = 1$

2) $C = -3$, $UB = -3 + 1 = -2$
 $LB = -3 - 1 = -4$

3) $P = \frac{2\pi}{1} = 2\pi$

4) $B = 0$



State the domain and range.

Domain: $(-\infty, \infty)$

Range: $[-4, -2]$

2) Graph the function.

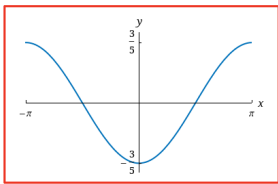
$$f(x) = -\frac{3}{5} \cos(x) \quad \text{LMHML}$$

1) $A = -\frac{3}{5}$

2) $C = 0$, $UB = 0 + (-\frac{3}{5}) = -\frac{3}{5}$
 $LB = 0 - (-\frac{3}{5}) = \frac{3}{5}$

3) $P = \frac{2\pi}{-\frac{3}{5}} = -\frac{10\pi}{3}$

4) $B = 0$



State the domain and range.

Domain: $(-\infty, \infty)$

Range: $[-\frac{3}{5}, \frac{3}{5}]$

3) Graph the function.

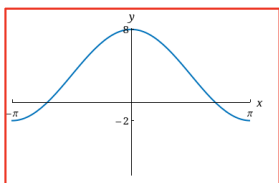
$$f(x) = 3 + 5 \cos(x) \quad \text{HMLMH}$$

1) $A = 5$

2) $C = 3$, $UB = 3 + 5 = 8$
 $LB = 3 - 5 = -2$

3) $P = \frac{2\pi}{5}$

4) $B = 0$



State the domain and range.

Domain: $(-\infty, \infty)$

Range: $[-2, 8]$

- 4) Find the amplitude and period of the function.

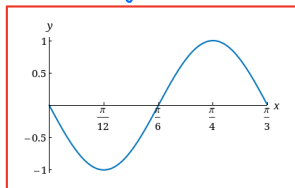
$$y = -\sin(6x)$$

$$= -\sin(6(x))$$

$$\text{Amplitude: } \boxed{1}$$

$$\text{period: } \frac{2\pi}{6} = \boxed{\frac{\pi}{3}}$$

Sketch the graph of the function.



$$c=0$$

$$b=0$$

MLMHM

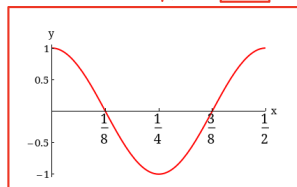
- 5) Find the amplitude and period of the function.

$$y = \cos(4\pi x)$$

$$y = \cos(4\pi(x))$$

$$\text{amplitude: } \boxed{1}$$

$$\text{period: } \frac{2\pi}{4\pi} = \boxed{\frac{1}{2}}$$



$$c=0$$

$$b=0$$

HMLMH

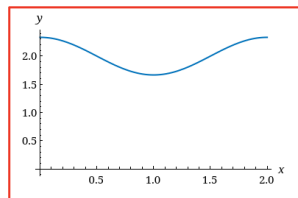
- 6) Find the amplitude and period of the function.

$$y = 2 + \frac{1}{3}\cos(\pi x)$$

$$y = 2 + \frac{1}{3}\cos(\pi(x))$$

$$\text{Amplitude: } \boxed{\frac{1}{3}}$$

$$\text{Period: } \frac{2\pi}{\pi} = \boxed{2}$$



HMLMH

$$c=2$$

$$b=0$$

- 7) Find the amplitude, period, and horizontal shift of the function.

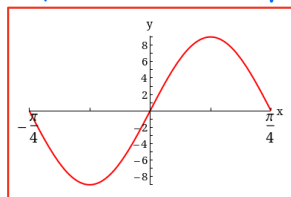
$$y = -9\sin\left(4\left(x + \frac{\pi}{4}\right)\right)$$

$$\text{Amplitude: } \boxed{9}$$

$$\text{Period: } \frac{2\pi}{4} = \boxed{\frac{\pi}{2}}$$

$$\text{horizontal shift: } \boxed{-\frac{\pi}{4}}$$

Graph one complete period.



MLMHM

- 8) Find the amplitude, period, and horizontal shift of the function.

$$y = 1 + \cos\left(4\left(x + \frac{\pi}{2}\right)\right)$$

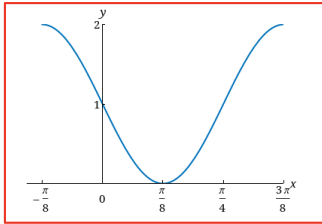
$$y = 1 + \cos\left(4\left(x + \frac{\pi}{2}\right)\right)$$

$$\text{Amplitude: } \boxed{1}$$

$$\text{Period: } \frac{2\pi}{4} = \boxed{\frac{\pi}{2}}$$

$$\text{Horizontal Shift: } \boxed{-\frac{\pi}{2}}$$

Graph one complete period.



HMLMH

- 9) Find the amplitude, period, and horizontal shift of the function.

$$y = \sin(\pi + 6x)$$

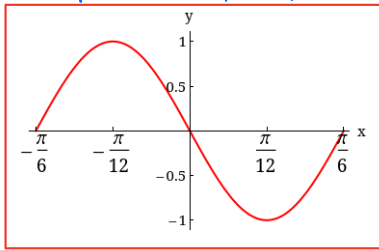
$$y = \sin\left(6\left(x + \frac{\pi}{6}\right)\right)$$

Amplitude: $\boxed{1}$

$$\text{Period: } \frac{2\pi}{6} = \boxed{\frac{\pi}{3}}$$

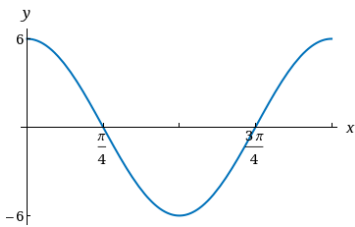
$$\text{Horizontal shift: } \boxed{-\frac{\pi}{6}}$$

Graph one complete period.



M+MLM

- 10) The graph of one complete period of a cosine curve is given.



Find the amplitude and period.

Amplitude: $\boxed{6}$

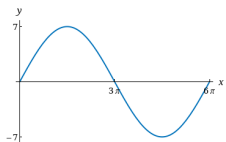
Period: $\boxed{\pi}$

$$\text{Equation: } \boxed{y = 6 \cos 2(x)} \quad \because P = \frac{2\pi}{|k|}$$

$$\pi = \frac{2\pi}{|k|}$$

$$|k| = 2$$

- 11) The graph of one complete curve of a sine function is given.



Find amplitude and period.

Amplitude: $\boxed{7}$

Period: $\boxed{6\pi}$

$$\text{Equation: } y = 7 \sin\left(\frac{\pi}{3}(x)\right) \quad \because P = \frac{2\pi}{|k|}$$

$$6\pi = \frac{2\pi}{|k|}$$

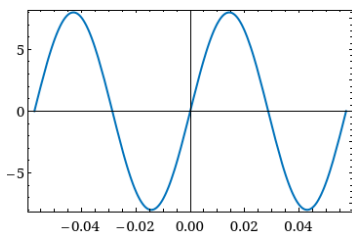
$$|k| = \boxed{\frac{1}{3}}$$

- 12) Use graphing calculator and determine appropriate viewing rectangle to draw graph.

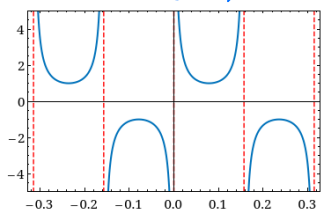
$$f(x) = 8 \sin(110x)$$

min & max: -8 and 8

$$\text{period: } \frac{2\pi}{110}$$



- 13) Use graphing calculator and determine appropriate viewing rectangle to draw graph.
 $f(x) = \csc(20x)$



min: 5
 max: -5
 period: $\frac{2\pi}{20}$

- 14) Use graphing calculator. Find all solutions that lie on the interval $[0, \pi]$
 $\tan(x) = 2$
 $x = 1.11$

- 15) A tuning fork is struck, producing a pure tone as its tines vibrate. The vibrations can be modeled by the function
 $v(t) = 0.8 \sin(800\pi t)$
 $v(t)$ is displacement of tines in millimeters at t seconds

(a) Find period in seconds of vibration

$$p = \frac{2\pi}{800\pi}$$

$$= \boxed{\frac{1}{400} \text{ s}}$$

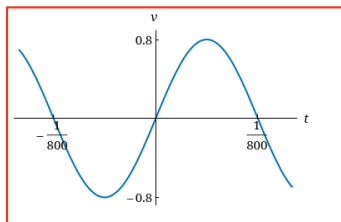
(b) frequency of vibrations

$$f = \frac{1}{p}$$

$$= \frac{1}{\frac{1}{400}}$$

$$= \boxed{400 \text{ vibrations/second}}$$

Graph



- 16) Find amplitude, period, and horizontal shift

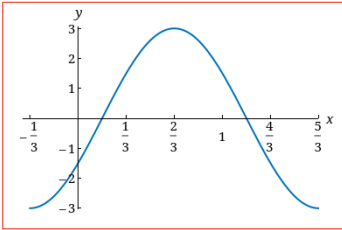
$$y = -3 \cos\left(\pi\left(x + \frac{1}{3}\right)\right)$$

amplitude: $\boxed{3}$

period: $\frac{2\pi}{\pi} = \boxed{2}$

horizontal shift: $\boxed{-\frac{1}{3}}$

Graph



LMHML